Ride-Hailing Demand & Smart Charging for an Electric Fleet

Advanced Analytics and Applications

Team 04

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1 Problem description

Urban mobility is in the middle of a structural shift. Over the past decade, ride-hailing has grown from a niche offering into a force that is disruptively reshaping urban transport (Wang & Yang, 2019), changing travel behaviour and, in a number of cities, drawing riders away from public transit (Tirachini, 2020). At the same time, road transport is electrifying rapidly: extrapolating historical sales with a logistic growth model, electric vehicles are projected to reach roughly a third of the global passenger-car stock by the early 2030s (Rietmann et al., 2020). These two trends are converging, and established car manufacturers increasingly seek to move beyond selling vehicles towards operating mobility services themselves. This is where our **business goal** begins: a german car company is entering the space with a ride-hailing platform run on a fully electrified fleet, targeting the more receptive US market. Building vehicles and profitably operating a fleet of them are distinct competencies, and running an on-demand service requires a quantitative understanding of where and when demand arises, which the manufacturer does not yet possess.

That understanding matters because demand for on-demand mobility has pronounced spatial and temporal structure (Ke et al., 2017) and is further shaped by exogenous conditions such as weather. An operator must therefore decide on three levels. Strategically, which districts of a city to enter first; tactically, how many vehicles to deploy and at which times and places; and operationally, how to position and, since the fleet is electric, charge those vehicles efficiently. Answering these questions requires demand data at fine spatial and temporal resolution, which does not yet exist for a platform that has not launched. As a proxy, we use the public taxi trip records of the City of Chicago from 2024 onwards: taxis are a reasonable stand-in for on-demand ride-hailing, and Chicago serves as a representative large US city. Nevertheless, the proxy is imperfect, as taxis differ from app-based hailing in booking behaviour and market coverage, a limitation we keep in view when drawing conclusions.

From this business problem we derive our **data-mining goal**, a chain of five analytics tasks that build on one another. First, *data preparation* turns the large, privacy-restricted raw trip records into demand panels: we validate and clean the records, discarding implausible or inconsistent trips, discretise the city with a hexagonal H3 grid, and enrich each cell and time window with external weather data. Second, *descriptive spatial analytics* characterises how demand varies across space and time, selects an appropriate spatial and temporal resolution, and identifies persistent demand hotspots using Gaussian Mixture Models and kernel density estimation. Third, *predictive analytics* estimates the expected demand per spatial unit and time window as a function of calendar, weather and location, using Support Vector Machines and feed-forward neural networks; the aim is conditional estimation, quantifying how demand depends on observable conditions, rather than autoregressive forecasting of the next interval from the current one. Fourth, *prescriptive analytics* addresses the electric dimension by designing a home-charging agent through reinforcement learning that minimises recharging cost while ensuring enough energy for the working day. Finally, *synthesis* consolidates these results into recommendations for the operator, including whether it should rely on private or public charging infrastructure. Together, the five tasks move from describing demand, to explaining it, to acting on it.

2 Data description

We draw on three datasets. The primary source is the *Taxi Trips (2024-)* record published by the City of Chicago on its open data portal, holding 14,802,849 trips across 23 fields. It spans 1 January 2024 to 31 March 2026 and covers 3,735 vehicles, 46 licensed taxi companies and all 77 of Chicago's community areas. To this we add historical weather for Chicago from the Open-Meteo API, at hourly resolution (19,728 observations of temperature, precipitation, rain, snowfall, wind speed and a categorical WMO weather code) alongside 822 daily aggregates that add sunshine duration and the number of precipitation hours. A GeoJSON of the 77 community-area boundaries supplies the geometry used to map and aggregate trips spatially. We decided against point-of-interest extensions, as the records already at hand cover the questions set out in Section 1, so the expected analytical gain did not justify the additional integration effort.

Each row represents one completed taxi trip. Location is reported at three nested levels, census tract within community area, plus a centroid coordinate pair. The 23 fields group as follows, with the full field-by-field dictionary given in the Appendix; for demand modelling only the pickup-side time and location fields are required.

Table 1: Structure of the raw Chicago taxi trip records: the 23 fields grouped by type, with the defining characteristics of each group.

Group	Fields	Notes
Identifiers	Trip ID, Taxi ID, Company	3,735 vehicles, 46 companies
Time	Trip Start / Trip End Timestamp	rounded to 15-minute bins
Trip metrics	Trip Seconds, Trip Miles	duration and distance
Space	Pickup / Dropoff Census Tract, Community Area, Centroid Latitude, Longitude and Location	coordinates are area centroids, not exact points
Money	Fare, Tips, Tolls, Extras, Trip Total	0.21% missing
Payment	Payment Type	

Two structural properties shape everything that follows. First, all timestamps are rounded to quarter-hour bins, taking only the minute values 0, 15, 30 and 45, which sets a hard floor on the temporal resolution any analysis can achieve. Second, location is restricted for privacy rather than by measurement: no exact coordinates are published, and the coordinate fields give the centroid of the enclosing area. Tract identifiers are further suppressed for the majority of trips, missing in 55.65% of pickups and 56.98% of dropoffs, whereas the community area is far more complete, missing in 2.79% and 8.92% of cases. Because suppression applies wherever a combination of tract and time holds too few trips to guarantee anonymity, it removes precisely the sparse observations and is therefore correlated with demand density. Analysis at tract level would be biased towards dense areas, which is why the choice of spatial unit in the next section cannot simply default to the finest available geography.

Two limitations remain. The raw records contain recording artefacts, most visibly trips of zero distance or zero duration, which we quantify and address in the next section. More fundamentally, the dataset captures served demand only: requests that went unmet leave no trace, so the data understate true demand in exactly those places and periods where supply was scarce.

3 Data preparation

The raw records are turned into an analysable demand panel in three stages: cleaning, discretisation and aggregation. Syntactic cleaning harmonises data types and parses timestamps, and no fully duplicated rows occur. Semantic validation then distinguishes two kinds of implausibility, and that distinction carries the whole preparation. Records that cannot describe a real journey are excluded outright: 98 trips end before they start, 12,104 last longer than three hours, and 1,202 exceed 100 miles. Records implausible in only one attribute are retained as demand events but withheld from the statistics they would distort, namely 1,294,820 trips of zero distance, 201,927 of zero duration, and fares above 200 dollars. A zero-distance record still marks a real pickup: deleting it would understate demand, while averaging it would corrupt the distance statistics, so it counts towards demand and is suppressed only where it would mislead. The complete set of validation rules and thresholds is documented in the Appendix.

Missing values are likewise not treated globally. Since the demand model describes where trips begin, missing dropoff information never removes an observation. Each derived dataset then carries its own requirement, so every analysis runs on the largest subset valid for it: the hexagon panel needs pickup coordinates, while the community-area panel, which we retain for the human-readable views in Section 4, needs the community area. Requiring pickup coordinates removes 405,819 trips, which together with the hard exclusions leaves 14,382,862 of 14,802,849 records, or 97.2%. We checked whether this distorts the temporal picture. Across months the missing rate is stable between 2.4% and 3.0%, but across the day it is not: it falls to 1.9% in the late afternoon and rises to 6.8% around 4 a.m. This is the density-driven mechanism from Section 2, as fewer night-time trips push more area and time combinations below the anonymity threshold. The effect is small in absolute terms, but night-time demand is marginally understated, which we revisit when interpreting results.

For the spatial unit we compared Chicago's community areas with a hexagonal grid. Community areas are administratively meaningful and human-readable, which makes them valuable for interpretation, but they vary considerably in size and shape, so demand densities are not comparable across them. We therefore discretise the city with the H3 indexing system, developed by Uber for exactly this purpose, which provides cells of uniform size and shape at a tunable resolution. We construct resolutions r4 to r8, where a higher index means smaller cells and each step divides the area by roughly seven. Two independent criteria bound the sensible range from both sides, as Table 2 shows. Downwards, r4 and r5 leave only 4 and 8 cells for the entire city, three of which already carry 80% of all demand, far too coarse to guide where vehicles should be deployed. Upwards, two things break at once. Sparsity rises steeply, with the share of empty cell and window combinations climbing from 12.6% at r6 to 49.2% at r7 and 82.0% at r8, so the panel increasingly records absence rather than demand. More fundamentally, measurement validity fails: because coordinates are area centroids of uneven precision, with trips of known tract carrying one of 607 tract centroids and suppressed trips one of merely 77 community-area centroids, a hexagon smaller than its reporting area cannot observe real structure. At r8, 67% of occupied hexagons are fed by a single centroid, so their apparent detail only reproduces where centroids happen to lie. Resolution r6 is the finest level at which both criteria remain intact, at more than three times the spatial detail of r5, and we adopt it as the canonical unit.

Table 2: Spatial resolution sweep at fixed four-hour windows. Coarse resolutions leave too few cells to guide deployment, while from r7 onwards sparsity and measurement validity deteriorate together, identifying r6 as the canonical unit.

Resolution	**Cell area**	**Cells**	**Empty share**	**Single-centroid cells**
r4	1,770.3 km^2	4	1.8%	0%
r5	252.9 km^2	8	1.0%	0%
r6	36.1 km^2	27	12.6%	0%
r7	5.2 km^2	128	49.2%	20%

Resolution	Cell area	Cells	Empty share	Single-centroid cells
r8	0.7 km^2	475	82.0%	67%

The temporal grid is bounded from below by the quarter-hour rounding of the source timestamps. We build one-hour, four-hour, eight-hour and daily windows and adopt four hours as canonical, since hourly bins reproduce the same sparsity problem while daily bins average away the within-day pattern on which any deployment decision depends. The weather records are aggregated to these windows and joined on the time key, contributing temperature, rain, snowfall, total precipitation, wind speed and the categorical weather code, so that demand variation driven by conditions is not misattributed to the calendar.

The aggregation returns one row per hexagon and time window, carrying the number of pickups as the target variable alongside mean trip distance, duration and fare, and the number of distinct taxis active in the cell. The window timestamp is retained as the basis for the cyclical calendar features, namely time of day, day of week and month, that enter the models in Section 4. Separately, we derive idle time per vehicle as the gap between one trip ending and the same taxi starting its next, which supports the utilisation view in Section 4. One step deserves emphasis. A grouped aggregation produces rows only where at least one trip occurred, so a cell without pickups is absent rather than zero, and a model trained on that panel would never observe the quiet half of the distribution and would systematically overestimate demand. We therefore complete the panel across the full Cartesian product of cells and windows and set the missing combinations to zero, leaving trip characteristics undefined because no trips exist to average. The canonical dataset comprises 27 hexagons across 4,927 four-hour windows, that is 133,029 observations, of which 12.6% are genuine zeros introduced by this step alone. Figure 1 makes the trade-off visible.

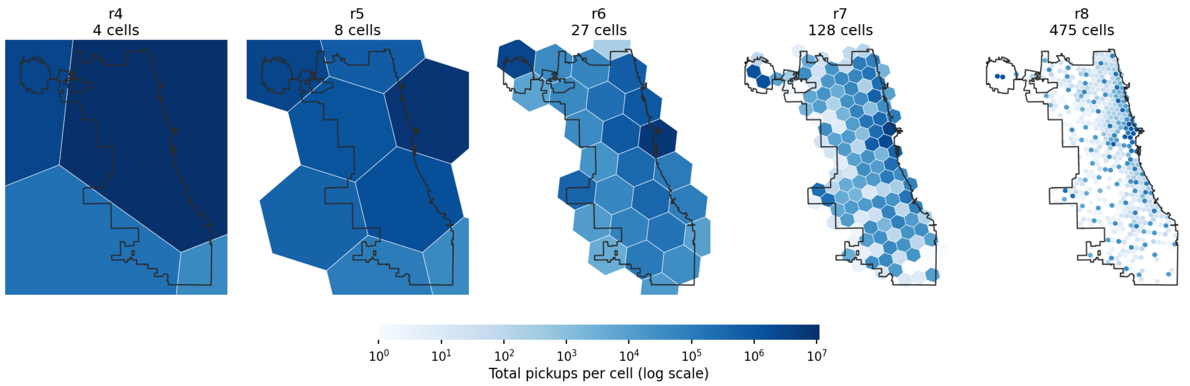

Figure 1: Total pickups per cell across H3 resolutions r4 to r8, on a shared logarithmic colour scale, with the city boundary for orientation. Coarse grids collapse the city into a handful of cells, while at r7 and r8 the surface fragments into largely empty ones. Resolution r6 is the finest level that still resolves spatial variation without dissolving into noise.

4 Data analytics

4.1 Descriptive Analytics

In terms of temporal dimension, city-wide demand follows a two-peak daily pattern, with the strongest peak around 5pm. Weekdays are busier than weekends, while demand also varies seasonally and weather adds a smaller secondary effect (See appendix Figure 2 and Figure 4). Warmer weather and rain both lift demand, while heavy snow suppresses it. This effect is real even after controlling for time of day (See appendix Figure 7, Figure 8, Figure 9).

A small number of areas (the Loop and Near North downtown core, and the two airports O'Hare and Midway) account for a large share of all pickups. The downtown core produces many short, high-throughput trips, whereas the airports produce fewer but longer and higher-value trips. Origin-destination flows are asymmetric and idle time is highest in low-demand periods and peripheral areas. These patterns are consistent with commuting, business activity, nightlife, and airport travel, but remain descriptive rather than causal findings.

The GMM and KDE show strong concentrations around central Chicago and the airport areas (See Figure 5 and Figure 6). Across resolutions, coarse spatial or daily aggregation loses operational detail, while fine hexagons and hourly bins produce many empty observations (See Figure 3). H3 r6 x 4h provides the best compromise between operational detail and sparsity and is therefore carried into predictive modelling.

4.2 Predictive Analytics

4.2.1 Modeling dataset and feature design

This section constructs two supervised models that predict taxi trip demand per spatial unit and time bucket in spatio-temporal resolution. The canonical modeling dataset is the **H3 r6 x 4h aggregation**. The prediction target is `demand_count`, the number of trips originating in a spatial unit during a time bucket. The panel is zero-filled, so the absence of trips is represented as an observation of zero rather than a missing row; this is required for the model to learn low-demand conditions rather than only observed trips. Demand is heavily right-skewed with 12.6% zeros, so it is modelled on a `log1p` scale and predictions are inverted before scoring.

Regarding independent variables, we conduct feature engineering process with these steps:

Calendar encoding. Hour-of-day, day-of-week, and month are periodic quantities, so each is encoded as a sine/cosine pair rather than a raw integer. This preserves continuity across the period boundary (23:00 is adjacent to 00:00, December to January), which an ordinal encoding would misrepresent as maximally distant. A binary weekend indicator is added because demand shifts between commuting and leisure regimes that are not fully captured by the day-of-week cycle alone.

Spatial encoding. Each spatial unit is represented by its centroid latitude and longitude rather than by a one-hot indicator per cell. Continuous coordinates encode relative position and distance, allowing the support vector machine and neural network to share information between neighbouring units and to generalise across space — which directly supports location-conditioned queries such as demand "around the outskirts". The centroid is derived deterministically from the H3 cell identifier, so reframing the retained spatial label as coordinates introduces no information loss.

Hotspot distances. The previous section identified three dominant demand centres: the downtown core, O'Hare, and Midway airports. The great-circle (Haversine) distance from each unit to these three anchors is added as an explicit feature. These distances are deterministic transforms of the centroid coordinates and therefore add no new information in principle, but they supply a coordinate frame aligned with the observed demand structure, which finite-sample SVM and neural-network models exploit more readily than raw coordinates. Three additional columns impose a negligible dimensional cost. The anchors are identified from training data only and are not hard-coded. Distances are expressed in miles for consistency with the trip-level distance fields.

Weather data. The weather features are aggregated into 4-hour bins from weather data, aligned with the demand time buckets. We use the mean for temperature and wind speed, the sum for rain, snow, and total precipitation, and the maximum weather code to capture the most severe condition within each window.

4.2.2 Validation strategy

In terms of train/test splitting, we split by grouping by calendar day, which means a day appears entirely in training data or testing data, never both. This is because the 4-hour buckets of a given day and cell share weather, calendar values, and unobserved local shocks. A purely random split could place some of a day's buckets in training and the rest in test, letting the model exploit day-specific signal absent from the features and inflating the score.

For feature standardization and hotspot anchor creation, we fit exclusively on training data. Standardization and the target transform are wrapped in a pipeline, ensuring they are automatically refit within each cross-validation fold.

For hyperparameter tuning and model selection, the NN uses five-fold GroupKFold over the development set. The linear SVM uses three-fold cross-validation, while the computationally expensive RBF-SVM uses three-fold GroupKFold on a 120-day development subsample. For testing and model comparison, a grouped random sample of whole days (15%) is set aside as an untouched test set.

To evaluate model performance, we use three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. MAE reports the average absolute deviation between predicted and observed demand in the original unit (trips per spatio-temporal bin), making it directly interpretable for non-technical stakeholders. RMSE penalizes larger errors more heavily due to the squaring term, making it more sensitive if the model underestimates demand during a spike. R^2 indicates the proportion of variance in demand explained by the model relative to a naive mean-baseline, giving a scale-free measure of overall fit.

4.2.3 Support vector machines

First, a linear support vector regressor is fitted as a simple baseline. A small grid search selects `C = 0.1` and `epsilon = 0.2`. `C` defines the strong penalty on model complexity while epsilon defines a tolerance margin around each prediction within which errors are ignored, so the model is penalized only for deviations larger than this margin. On the holdout set, the linear SVM achieves MAE 65.56, RMSE 203.49, and R^2 0.633. As a linear model, it cannot capture nonlinear interactions between location and time, which contributes to its larger errors during demand peaks.

Second, an RBF kernel is introduced to capture nonlinear relationships between time, location, and weather. The three-fold grouped grid search is run on a random 120-day development subsample because kernel SVMs are computationally expensive. The selected parameters are `C = 10`, `epsilon = 0.1`, and `gamma = 0.1` (controlling how far the influence of a single training point reaches). The RBF-SVM improves holdout performance to MAE 30.83, RMSE 118.86, and R^2 0.875.

Its main shortfalls are computational cost and limited scalability. The selected RBF estimator is not refitted on the full development set, takes approximately 18 seconds to score the holdout, and does not outperform the historical-mean benchmark (MAE 29.14, RMSE 107.13,

R^2 0.898), suggesting that within this spatio-temporal resolution, most predictable demand variation is already captured by historical averages, and the added model complexity yields limited marginal value.

4.2.4 Neural networks

We also use a feedforward neural network to predict taxi demand. Our model development process includes developing a baseline model and tuning its hyperparameters.

First, the feedforward baseline is fitted without hyperparameter tuning. It uses the same feature matrix, splits, and scaling as the SVM models. Its architecture consists of an input layer sized to the feature count, two hidden layers with 64 and 32 nodes, ReLU activations, and an output layer (see Appendix). Other settings include:

Dropout = 0.2. During training, 20% of hidden-layer activations are randomly set to zero. This regularises the network by reducing co-adaptation and dependence on individual activations.

Optimiser. Adam with lr=1e-3. By using Adam for optimiser, each parameter gets its own adaptive step size, scaled down where gradients are large or noisy, scaled up where they are small or consistent.

Early stopping. Training stops when validation loss fails to improve for 15 epochs and restores the weights from the best-performing epoch. The internal 10% validation split is drawn from the respective training fold and does not overlap with the outer validation fold or final holdout. The baseline result is shown in the Appendix.

Second, we vary the number and width of hidden layers, dropout, learning rate, and batch size. The candidate values shown in the Appendix are evaluated by manual grid search with five-fold grouped cross-validation.

Among the candidate configurations evaluated via grid search, the best-performing model was found with hidden sizes (128, 64, 32), dropout of 0.1, learning rate of 1e-3, and batch size of 512. The average `demand_count` of the dataset is 108. The MAE of the best model is 22, which is only 20% of average. It indicates that model is quite good in predicting the demand. Meanwhile, RMSE is much more larger than MAE. It may be because the model makes huge mistakes with the spatial unit that have very high demand. R2 = 93.54. It means that model can explain 93.54% of variance. In other word, it shows that model is 93.54% than the model predicts all rows as mean of demand.

To see how the variables contribute to the prediction, we analyze SHAP values (SHapley Additive exPlanations), which indicate how much each feature pushed the prediction up (+) or down (-). The SHAP value of feature i is calculated by adding feature i to every possible subset of features, measuring how much the prediction changes, and averaging that change across all subsets with appropriate weights.

The SHAP plot can be found in the Appendix {}. The most important features are related to spatial unit, especially the **distance** from the hexagon to the busy areas. Low distance to Loop makes the demand increase, which means the hexagon near the busy zones have high demand. It aligns with the business sense. The **time features** (`hour_cos`, `hour_sin`) also have the high ranks, which means time of day matters a lot. `is_weekend` (weekend = 1) slighly negative. It means that weekdays have slightly higher taxi demand overall. The **weather features** do not have much contribution to the demand, the SHAP value is nearly 0.

4.2.5 Resolution sensitivity

To assess how model performance varies with spatial and temporal resolution, we re-evaluate model performance along two dimensions: (1) spatial resolution (H3 r5/r6/r7, Community Area, Census Tract) at fixed 4h bins, and (2) temporal resolution (1h, 4h, 8h, 1d) at fixed H3 r6. We use the same set of hyperparameters and model configurations as in 4.2.3 and 4.2.4 and only refit them on each resolution's training split. The hotspot anchors and the scaler are refitted per resolution because they are data-derived.

Regarding spatial sensitivity, the granularity–sparsity trade-off is clear. From r5 to r7, the number of cells grows from 8 to 128, mean demand falls from 357 to 22, and the zero-rate climbs from 1% to 49%. R^2 weakens accordingly. Raw MAE falls because finer cells hold fewer trips; hence, cross-resolution comparisons should rely on R^2/nMAE, not raw MAE. Community Area (77 cells, R^2 0.69–0.77) sits between r6 and r7. This result shows that finer resolution gives more localized signals but degrades reliability once cells become sparse. R6 (or Community Area) is the more defensible operating resolution for fleet decisions than r7, where the low R^2 indicates limited explanatory power. However, it should be noted that the per-resolution neural network is a single refit and underperforms its tuned version.

{Note: add result table in the Appendix}

Census tract has 607 unique cells, 94.7% of buckets empty, and R^2 collapses to 0.08 (SVM) / 0.18 (NN) - essentially no better than predicting the mean. Two effects compound. First, the grid is extremely fine, so sparsity alone depresses R^2, as it already did at r7. Second, only **45.6%** of trips carry a tract and the missing ones are the low-demand ones, so even the observed demand is a biased, deflated picture. The conclusion for the client is that census tract is unsuitable as a primary unit - not because the model is weak, but because the input is both too fine and structurally incomplete in a demand-correlated way.

{Note: add result table in the Appendix}

Regarding temporal sensitivity, from 1h to 1d the zero-rate falls from 26.8% to 4.0% and R^2 rises while raw MAE grows with the bigger counts. But high R^2 at 1d is not a free win: a one-day bucket cannot answer "how many cars at 8am vs 6pm", the tactical question the client actually has. 4h is enough temporal grain for shift planning, sparse enough to stay learnable.

{Note: add result table in the Appendix}

4.2.6 SVM vs. NN comparison

On the holdout set, the tuned NN is the best model (MAE 24.9, R2 0.921), the NN baseline is essentially as good (MAE 25.8, R2 0.926), and both clearly beat the RBF-SVM (MAE 30.8, R2 0.875) and the linear SVM (MAE 65.6). Three points decide the verdict:

- **The SVM does not beat the naive benchmark.** The historical-mean lookup scores MAE **29.1 / R2 0.898** - better than the RBF-SVM on both metrics. Only the NN (MAE 24.9 / R2 0.921) adds real signal over a trivial baseline. A kernel SVM that, after a full grid search, fails to beat a lookup table is hard to justify as the deployed model.
- **Cost runs the same way.** The RBF-SVM had to be tuned on a day-subsample (kernel SVMs scale ~quadratically in samples) and takes **~18 s** to score the holdout; the NN trains on the full development set and scores in **~0.01 s** - roughly 1800x faster inference. The deep model is both more accurate and cheaper to serve.
- **Interpretability** is not lost: SHAP (3.6) shows time-of-day and the hotspot distances dominate - enough for a client conversation.

Therefore, deep learning is clearly worth it here: the NN is the only model that meaningfully beats the benchmark, and it is also the faster one to serve.

{Note: add result table in the Appendix}

4.3 Smart Charging Using Reinforcement Learning

4.3.1 MDP and model setup

The assignment's two-hour home-charging window is divided into eight 15-minute decisions. The state contains the slot and battery charge; actions are 0, 7, 14, or 22 kW, and 60 kWh is chosen as the battery-capacity parameter. Energy demand is modelled as `max(0, D)`, where `D` is normal with mean 21.4 kWh and standard deviation 14.1 kWh. These parameters are derived from pooled taxi-day distances using an assumed conversion of 0.40 kWh/mile. Slot cost is `alpha_t * (exp(0.1 * power) - 1)`. A shortfall incurs 20 plus 9.54 per missing kWh. The first day starts at 5 kWh; as an extension to the single-day setup, later days start with the preceding day's remainder.

4.3.2 Learning approach

Tabular Q-learning uses the slot and battery charge, snapped to a 0.5 kWh grid, with the four charging actions. It is trained for 200,000 simulated days using epsilon-greedy exploration; epsilon decreases from 1.0 to 0.02 and the learning rate decays. Future rewards are undiscounted within a day and discounted by 0.95 at the day boundary. Dynamic programming (DP) uses the same action set and battery grid and converges after 187 iterations. However, DP also approximates demand on an 81-point grid that does not reproduce the simulator's clipped normal distribution exactly. DP is therefore an approximate model-based reference, not proof of the exact optimum in the simulation environment.

4.3.3 Results and comparison

All policies are evaluated for 5,000 days using the same random seed and hence the same demand sequence. The DP policy has the lowest average charging-cost value, 1.0521, and the second-lowest shortfall rate, 0.52%. Q-learning reaches 1.3191 and 3.02%. Constant-medium charging reaches 1.2844 and 3.90%, so, based on the reported values, Q-learning reduces shortfall by 0.88 percentage points for 0.0347 additional charging cost. Maximum charging has the lowest shortfall rate, 0.46%, but costs 3.3737; the cheapest-slots heuristic reaches 2.1106 and 1.44%. Thus, Q-learning improves reliability over constant-medium charging but does not dominate all baselines. Because the output omits penalty-inclusive return, it supports a cost-reliability comparison rather than a complete ranking by the training objective.

4.3.4 Implications and limitations

The sensitivity analysis re-solves DP for each setting and evaluates each resulting policy over 2,000 days. Increasing demand's standard deviation from 5 to 20 kWh raises average charging cost from 0.8056 to 1.3114 and shortfall from 0.10% to 3.60%. Raising both penalty components from 0.5 to 4 times their base values increases charging cost from 1.0460 to 1.1248 and reduces shortfall from 0.45% to 0.25%. Tested battery capacities of 40, 60, and 80 kWh produce observed shortfall rates of 9.20%, 0.15%, and 0.00%, respectively. These are model outcomes, not dollar costs or real-world forecasts. They depend on the chosen penalty, assumed energy conversion, clipped normal demand, stylised prices, fixed charging window, and omission of charging losses, public charging, battery degradation, and fleet interactions.

5 Discussion & Outlook

5.1 Implications for the fleet operator

For the client, the main implication is that market entry should start with a focused operating area rather than immediate city-wide coverage. The cleaned Chicago taxi data, enriched with weather data and aggregated to H3 r6 x 4h, shows that demand is strongly structured by time and location. Central areas and airport-related zones require different operating logic: dense downtown areas support frequent short trips and high turnover, while airport trips are longer, more range-sensitive, and operationally more demanding. Since taxis are only a proxy for ride-hailing demand, these findings should not be transferred blindly to another market, but the pipeline can be reused for the actual target city.

Strategically, the operator should first cover persistent demand centres and the corridors between them. Tactically, the tuned neural network is the preferred demand model because it performs best against the historical-mean benchmark, while H3 r6 x 4h offers a practical compromise between operational detail and sparsity. However, the prediction target is trip demand, not fleet size. Fleet requirements still depend on trip duration, idle time, vehicle productivity, target service levels, repositioning, and charging downtime.

5.2 Further analyses using the existing data

The most important next step is to translate predicted demand into vehicle requirements. The available taxi data already contains trip duration, idle time, and taxi identifiers, which could be used to estimate how many trips one active vehicle can realistically serve per area and time bucket. Combining this productivity estimate with predicted demand would provide a first fleet-sizing model, including reserve capacity for peaks and prediction errors.

A second extension is a simple fleet simulation. Origin-destination flows can be used to model where vehicles accumulate, where shortages occur, and how much empty repositioning is needed. This would connect the descriptive flow analysis with the predictive demand model and the smart-charging component. Instead of only asking where demand occurs, the operator could evaluate whether enough vehicles are available in the right places at the right times.

Further model validation should focus on robustness. Rolling time-based validation, prediction intervals, and residual analysis would show whether the model remains reliable across seasons, weather conditions, and unusual demand periods. This matters because operational planning depends not only on average performance, but also on how badly the model fails in peak situations.

5.3 Additional external data sources

Several external data sources could improve the analysis. Point-of-Interest data from OpenStreetMap would be especially useful, because offices, airports, transit stations, hotels, residential areas, and entertainment venues help explain why demand differs between spatial cells. Event calendars, public holidays, traffic speeds, road closures, and public-transport disruptions could further improve demand and travel-time predictions.

For the charging part, the current model should be extended with real vehicle and infrastructure data. Relevant sources include vehicle telematics, battery state-of-charge records, energy consumption by trip type, home or depot availability, electricity tariffs, grid constraints, and public charger data. Public charger data should include location, power level, price, utilisation, waiting times, and outages. Without this information, the charging model should be interpreted as a controlled proof of concept rather than a final infrastructure decision.

5.4 Charging infrastructure recommendation

The results support a hybrid charging strategy. Private charging should form the operational backbone because it offers more control over timing, availability, and cost. The smart-charging model shows that scheduled charging can maintain high reliability without always using maximum power. This is especially valuable for vehicles with predictable home or depot charging windows.

Public charging should complement private charging rather than replace it. It is useful as a reserve option for unexpectedly high energy demand, low starting charge, vehicles operating far away from private chargers, or peak-demand days where normal charging windows are insufficient. However, public charging introduces uncertainty through detours, waiting times, charger availability, and price variation, none of which are fully captured in the current model.

Therefore, the client should not decide immediately between fully private or fully public charging. A sensible next step is a pilot in the target market. The pilot should collect local ride-hailing demand, vehicle utilisation, energy consumption, charging behaviour, and public charging reliability data. Based on this evidence, the operator can size a private charging backbone and define how much public charging capacity is needed as flexible backup.

6 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like "R for Data Science" (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

7 Conclusion

The conclusion should summarize your findings and suggest future work.

8 Appendix

8.1 H3 hexagon area corresponding to specific resolution

Resolution	Avg Area	Rough Scale
R4	~1,770 km^2	Large city / metro region
R5	~253 km^2	City district
R6	~36 km^2	Large neighborhood
R7	~5.2 km^2	Neighborhood
R8	~0.74 km^2	A few city blocks

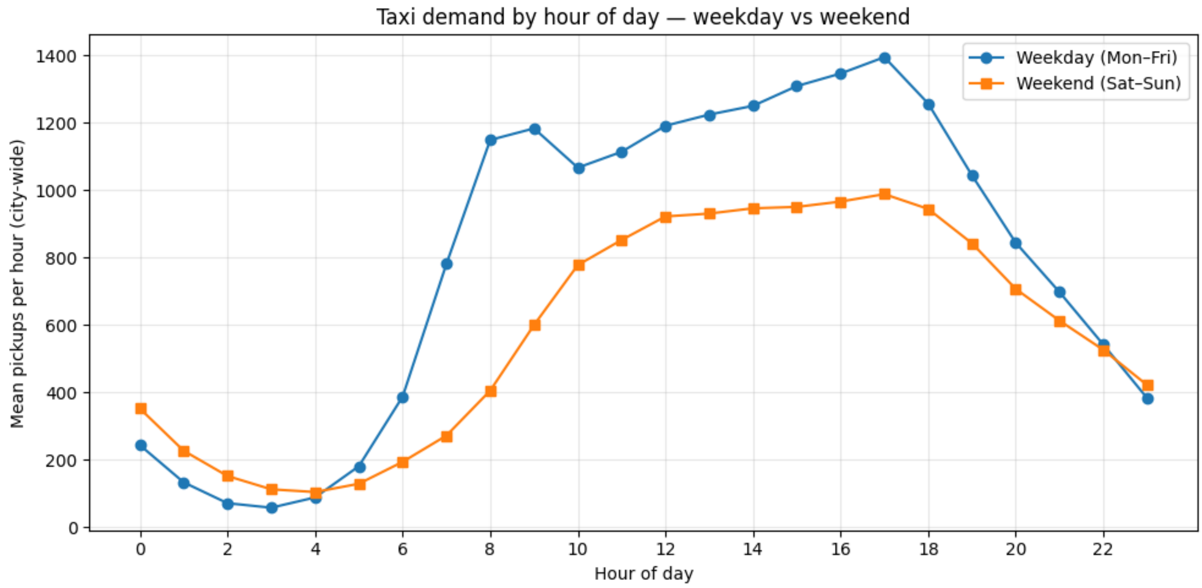

Figure 2: Taxi demand by hour of day - weekday vs weekend

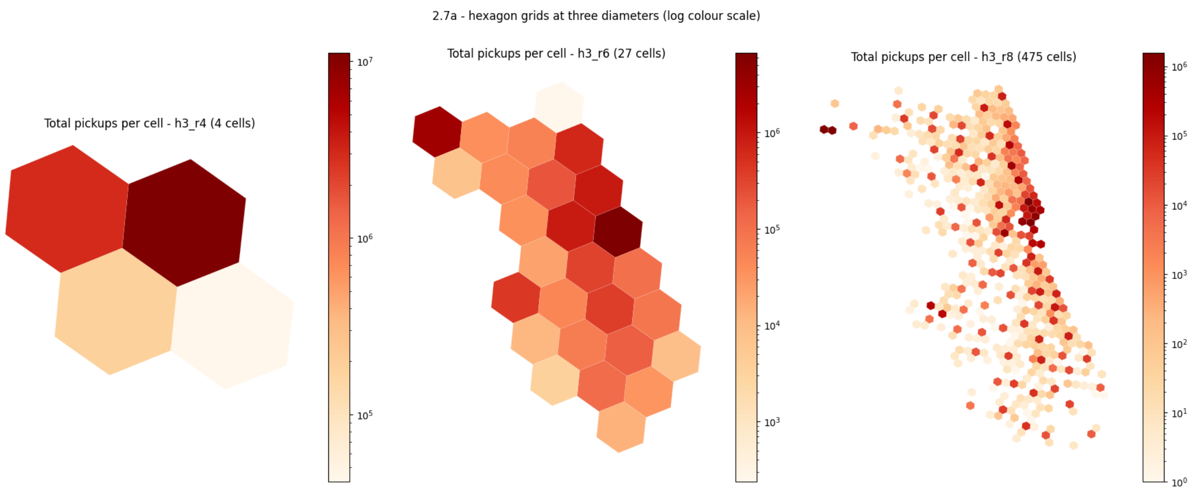

Figure 3: Total pickups per cell

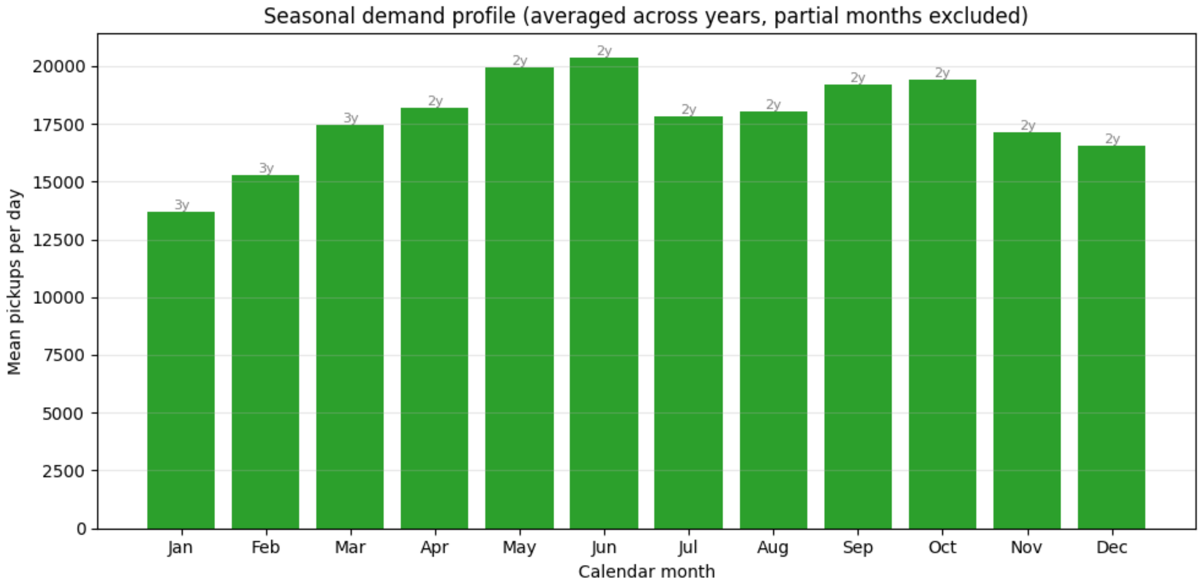

Figure 4: Seasonal demand profile

8.2 Baseline Neural Network Architecture

8.3 candidate values of hyperparameters for tuning Neural Network model

Parameter	Candidate Values
`hidden_sizes`	(64, 32), (128, 64, 32)
`dropout`	0.1, 0.2
`lr`	1e-3, 5e-4
`batch_size`	256, 512

8.4 Baseline Neural Network model result

8.5 Best Neural Network model

8.6 Feature importance - SHAP - of best Neural Network model

References

Ke, J., Zheng, H., Yang, H., & Chen, X. ((2017). Short-term forecasting of passenger demand under on-demand ride services: A spatio-temporal deep learning approach. *Transportation Research Part C: Emerging Technologies*, *85*, 591–608.

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Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science*. O'Reilly Media.

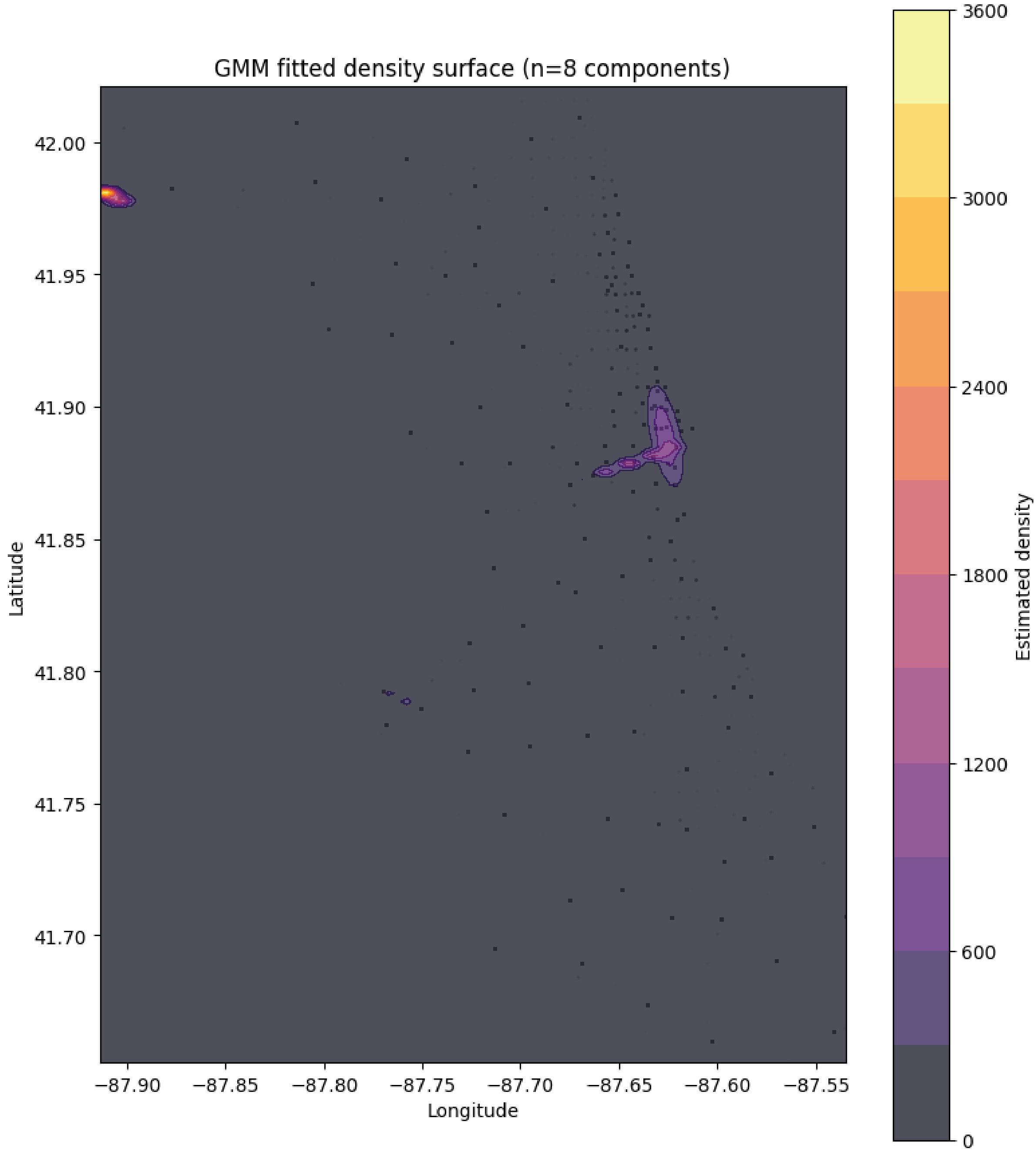
GMM fitted density surface (n=8 components)
Latitude
Longitude
Estimated density
42.00
41.95
41.90
41.85
41.80
41.75
41.70
−87.90
−87.85
−87.80
−87.75
−87.70
−87.65
−87.60
−87.55
3600
3000
2400
1800
1200
600
0

Figure 5: GMM

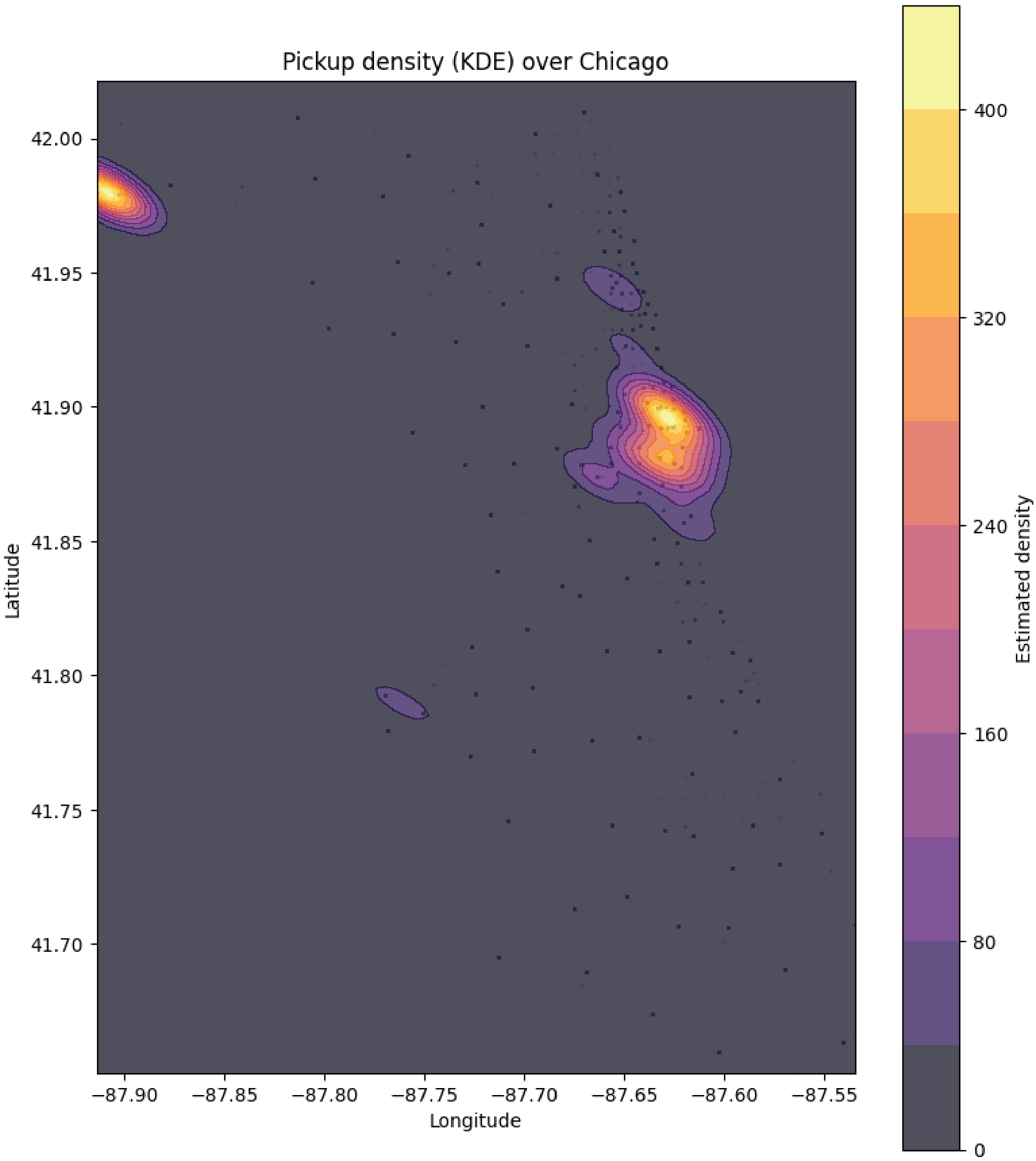
Pickup density (KDE) over Chicago
42.00
41.95
41.90
41.85
41.80
41.75
41.70
Latitude
−87.90
−87.85
−87.80
−87.75
−87.70
−87.65
−87.60
−87.55
Longitude
400
320
240
160
80
0
Estimated density

Figure 6: KDE

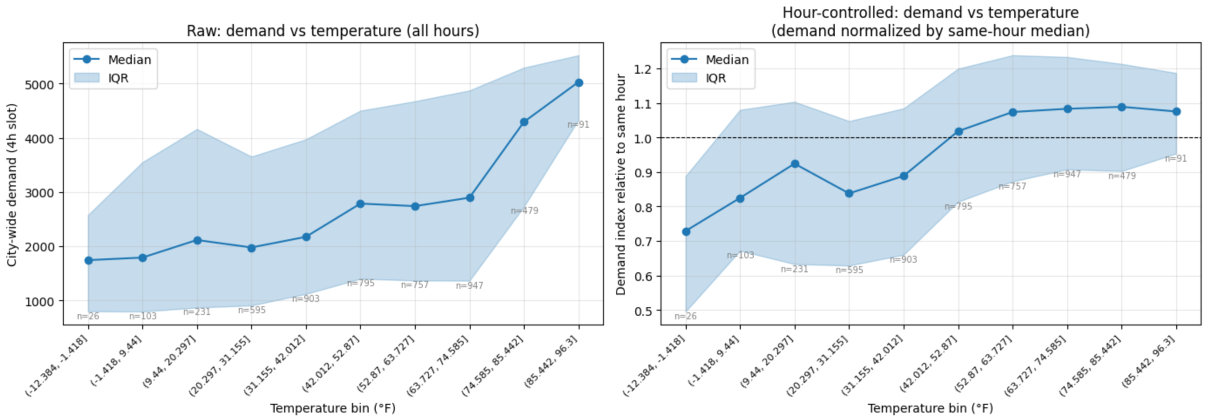

Figure 7: Demand vs temperature

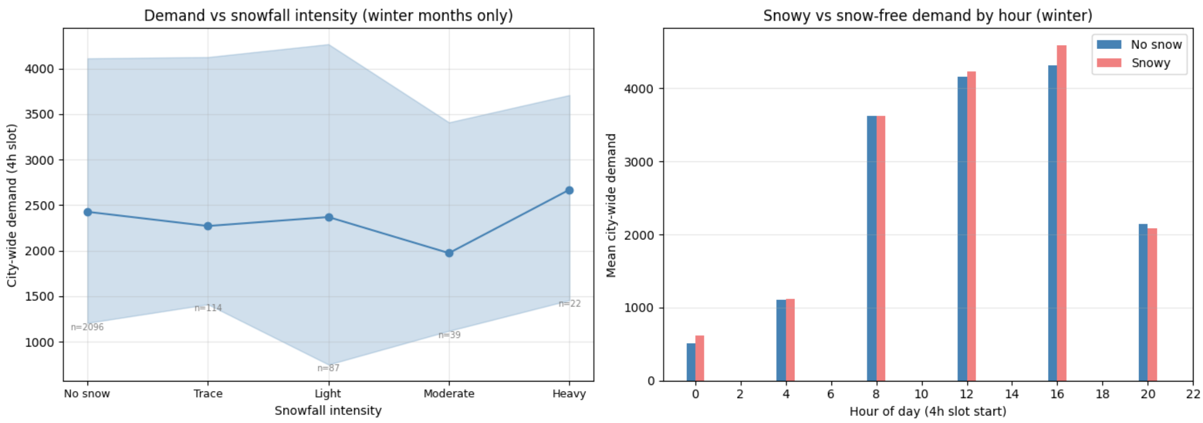

Figure 8: Demand vs snow intensity

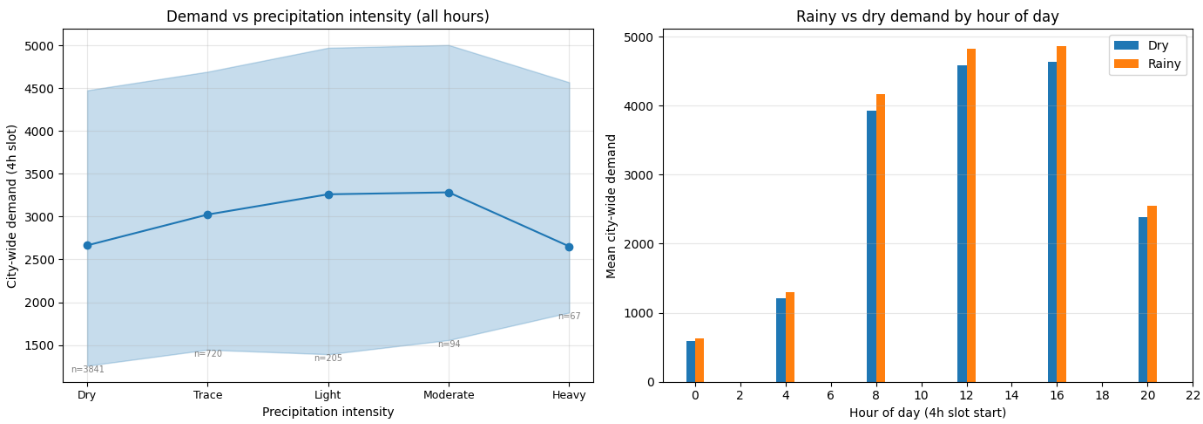

Figure 9: Demand vs precipitation intensity

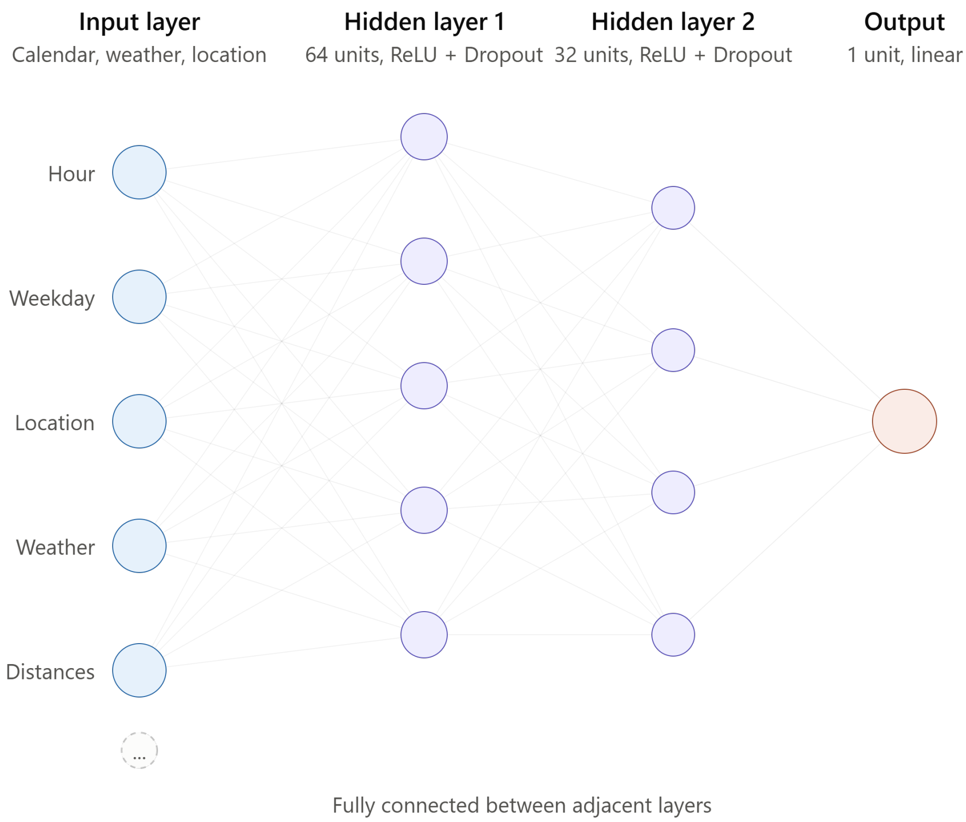

Figure 10: Model architecture

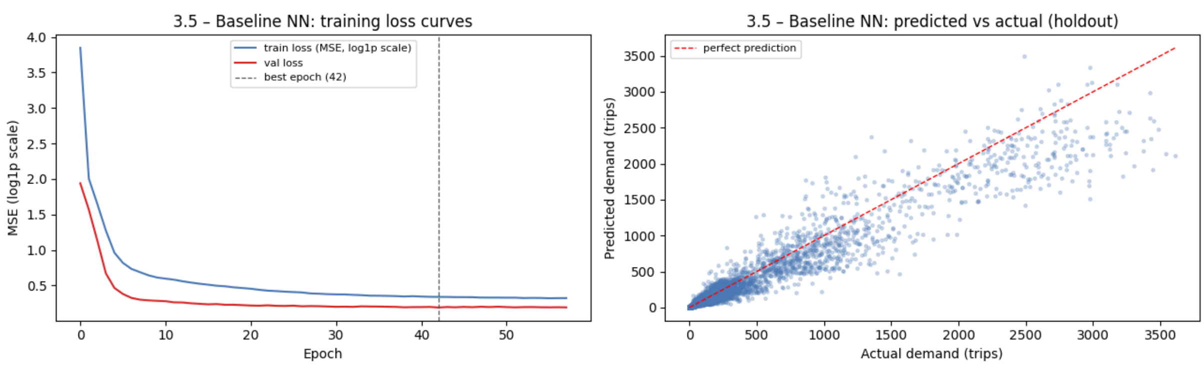

Figure 11: Model result - Neural network baseline model

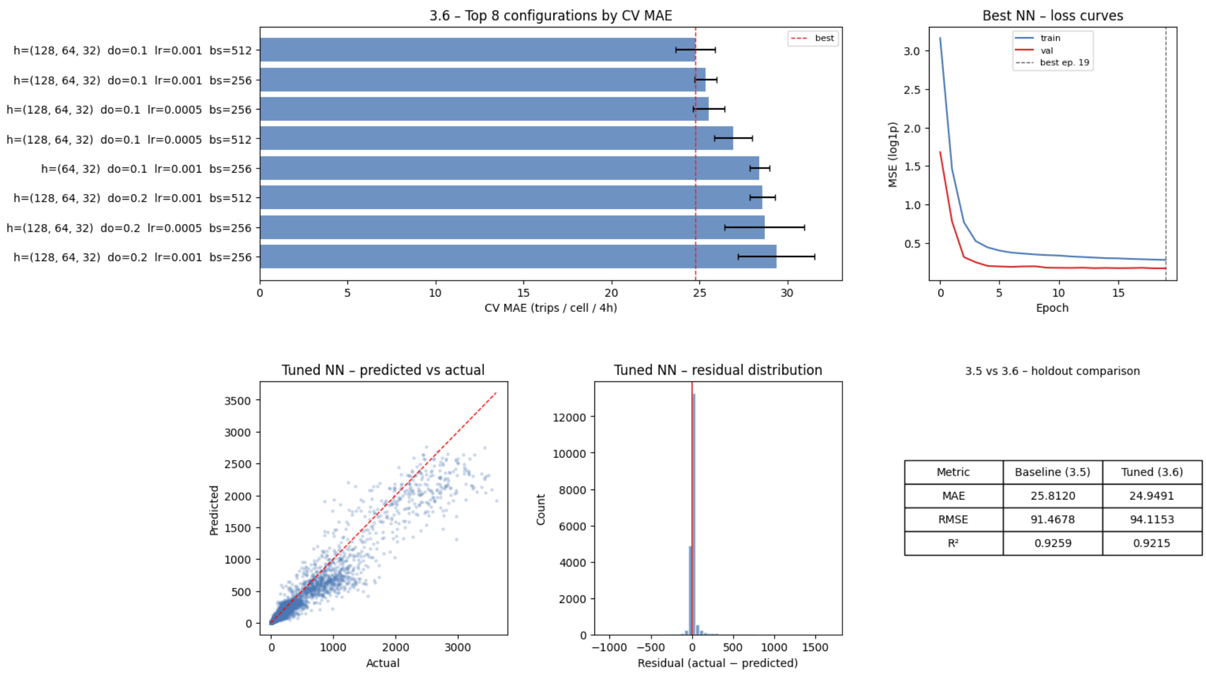

Figure 12: Model result - Neural network tuning model

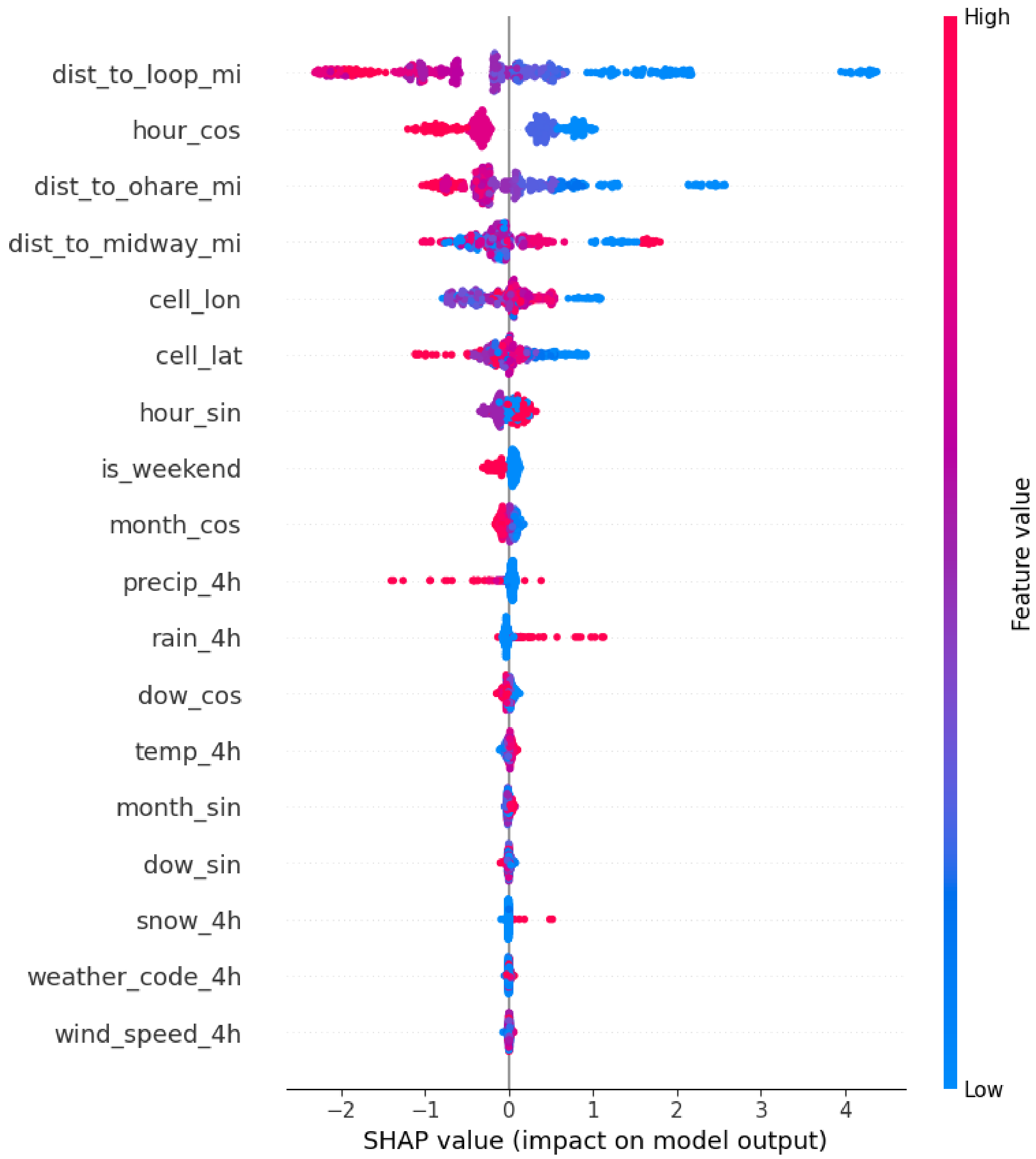

Figure 13: SHAP feature importance